$$C_{\phi d}^{(22)\,i1_i2_} \to$$

$$\frac{1}{16\,\pi^2}\Big(-\frac{2}{81}\sum_p g_1^4\,LF_{3,0}\big[m_{\tilde d}^{\,p}\big]\,\delta_{i1i2}+\frac{5}{108}\sum_p g_1^4\,LF_{4,-1}\big[m_{\tilde d}^{\,p}\big]\,\delta_{i1i2}-\frac{8}{405}\sum_p g_1^4\,LF_{5,-2}\big[m_{\tilde d}^{\,p}\big]\,\delta_{i1i2}-$$
$$\frac{2}{27}\sum_p g_1^4\,LF_{3,0}\big[m_{\tilde e}^{\,p}\big]\,\delta_{i1i2}+\frac{5}{36}\sum_p g_1^4\,LF_{4,-1}\big[m_{\tilde e}^{\,p}\big]\,\delta_{i1i2}-\frac{8}{135}\sum_p g_1^4\,LF_{5,-2}\big[m_{\tilde e}^{\,p}\big]\,\delta_{i1i2}-$$
$$\frac{1}{27}\sum_p g_1^4\,LF_{3,0}\big[m_{\tilde l}^{\,p}\big]\,\delta_{i1i2}+\frac{5}{72}\sum_p g_1^4\,LF_{4,-1}\big[m_{\tilde l}^{\,p}\big]\,\delta_{i1i2}-\frac{4}{135}\sum_p g_1^4\,LF_{5,-2}\big[m_{\tilde l}^{\,p}\big]\,\delta_{i1i2}-$$
$$\frac{1}{81}\sum_p g_1^4\,LF_{3,0}\big[m_{\tilde q}^{\,p}\big]\,\delta_{i1i2}+\frac{5}{216}\sum_p g_1^4\,LF_{4,-1}\big[m_{\tilde q}^{\,p}\big]\,\delta_{i1i2}-\frac{4}{405}\sum_p g_1^4\,LF_{5,-2}\big[m_{\tilde q}^{\,p}\big]\,\delta_{i1i2}-$$
$$\frac{8}{81}\sum_p g_1^4\,LF_{3,0}\big[m_{\tilde u}^{\,p}\big]\,\delta_{i1i2}+\frac{5}{27}\sum_p g_1^4\,LF_{4,-1}\big[m_{\tilde u}^{\,p}\big]\,\delta_{i1i2}-\frac{32}{405}\sum_p g_1^4\,LF_{5,-2}\big[m_{\tilde u}^{\,p}\big]\,\delta_{i1i2}-$$
$$\frac{1}{27}\,g_1^4\,LF_{3,0}[\tilde\mu]\,\delta_{i1i2}-\frac{1}{18}\,g_1^4\,LF_{4,-1}[\tilde\mu]\,\delta_{i1i2}+\frac{8}{135}\,g_1^4\,LF_{5,-2}[\tilde\mu]\,\delta_{i1i2}-$$
$$\frac{1}{324}\,g_1^4\,LF_{2,1,0}\big[m_1,m_{\tilde d}^{\,i1}\big]\,\delta_{i1i2}-\frac{1}{324}\,g_1^4\,LF_{2,2,-1}\big[m_1,m_{\tilde d}^{\,i1}\big]\,\delta_{i1i2}+$$
$$\frac{1}{162}\,g_1^4\,LF_{3,1,-1}\big[m_1,m_{\tilde d}^{\,i1}\big]\,\delta_{i1i2}-\frac{1}{324}\,g_1^4\,LF_{4,1,-2}\big[m_1,m_{\tilde d}^{\,i1}\big]\,\delta_{i1i2}-$$
$$\frac{1}{324}\,g_1^4\,LF_{2,1,0}\big[m_1,m_{\tilde d}^{\,i2}\big]\,\delta_{i1i2}-\frac{1}{324}\,g_1^4\,LF_{2,2,-1}\big[m_1,m_{\tilde d}^{\,i2}\big]\,\delta_{i1i2}+$$
$$\frac{1}{162}\,g_1^4\,LF_{3,1,-1}\big[m_1,m_{\tilde d}^{\,i2}\big]\,\delta_{i1i2}-\frac{1}{324}\,g_1^4\,LF_{4,1,-2}\big[m_1,m_{\tilde d}^{\,i2}\big]\,\delta_{i1i2}-$$
$$\frac{1}{36}\,g_1^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{2,1,0}\big[m_1,m_{\tilde q}^{\,p}\big]+\frac{1}{18}\,g_1^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{3,1,-1}\big[m_1,m_{\tilde q}^{\,p}\big]-$$
$$\frac{1}{36}\,g_1^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{4,1,-2}\big[m_1,m_{\tilde q}^{\,p}\big]-\frac{3}{4}\,g_2^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{2,1,0}\big[m_2,m_{\tilde q}^{\,p}\big]+$$
$$\frac{3}{2}\,g_2^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{3,1,-1}\big[m_2,m_{\tilde q}^{\,p}\big]-\frac{3}{4}\,g_2^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{4,1,-2}\big[m_2,m_{\tilde q}^{\,p}\big]-$$
$$\frac{1}{27}\,g_1^2\,g_3^2\,LF_{2,1,0}\big[m_3,m_{\tilde d}^{\,i1}\big]\,\delta_{i1i2}-\frac{1}{27}\,g_1^2\,g_3^2\,LF_{2,2,-1}\big[m_3,m_{\tilde d}^{\,i1}\big]\,\delta_{i1i2}+$$
$$\frac{2}{27}\,g_1^2\,g_3^2\,LF_{3,1,-1}\big[m_3,m_{\tilde d}^{\,i1}\big]\,\delta_{i1i2}-\frac{1}{27}\,g_1^2\,g_3^2\,LF_{4,1,-2}\big[m_3,m_{\tilde d}^{\,i1}\big]\,\delta_{i1i2}-$$
$$\frac{1}{27}\,g_1^2\,g_3^2\,LF_{2,1,0}\big[m_3,m_{\tilde d}^{\,i2}\big]\,\delta_{i1i2}-\frac{1}{27}\,g_1^2\,g_3^2\,LF_{2,2,-1}\big[m_3,m_{\tilde d}^{\,i2}\big]\,\delta_{i1i2}+$$
$$\frac{2}{27}\,g_1^2\,g_3^2\,LF_{3,1,-1}\big[m_3,m_{\tilde d}^{\,i2}\big]\,\delta_{i1i2}-\frac{1}{27}\,g_1^2\,g_3^2\,LF_{4,1,-2}\big[m_3,m_{\tilde d}^{\,i2}\big]\,\delta_{i1i2}-$$
$$\frac{4}{3}\,g_3^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{2,1,0}\big[m_3,m_{\tilde q}^{\,p}\big]+\frac{8}{3}\,g_3^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{3,1,-1}\big[m_3,m_{\tilde q}^{\,p}\big]-$$
$$\frac{4}{3}\,g_3^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{4,1,-2}\big[m_3,m_{\tilde q}^{\,p}\big]-\frac{1}{9}\,g_1^2\,\overline{a_d}^{pr}\,a_d^{\,pr}\,LF_{2,2,0}\big[m_{\tilde d}^{\,r},m_{\tilde q}^{\,p}\big]\,\delta_{i1i2}+$$
$$\frac{1}{18}\,g_1^2\,\overline{a_d}^{pr}\,a_d^{\,pr}\,LF_{3,2,-1}\big[m_{\tilde d}^{\,r},m_{\tilde q}^{\,p}\big]\,\delta_{i1i2}+\frac{1}{162}\,g_1^4\,LF_{2,1,0}\big[m_{\tilde d}^{\,i1},m_1\big]\,\delta_{i1i2}-$$
$$\frac{1}{324}\,g_1^4\,LF_{3,1,-1}\big[m_{\tilde d}^{\,i1},m_1\big]\,\delta_{i1i2}+\frac{2}{27}\,g_1^2\,g_3^2\,LF_{2,1,0}\big[m_{\tilde d}^{\,i1},m_3\big]\,\delta_{i1i2}-$$
$$\frac{1}{27}\,g_1^2\,g_3^2\,LF_{3,1,-1}\big[m_{\tilde d}^{\,i1},m_3\big]\,\delta_{i1i2}+\frac{1}{162}\,g_1^4\,LF_{2,1,0}\big[m_{\tilde d}^{\,i2},m_1\big]\,\delta_{i1i2}-$$
$$\frac{1}{324}\,g_1^4\,LF_{3,1,-1}\big[m_{\tilde d}^{\,i2},m_1\big]\,\delta_{i1i2}+\frac{2}{27}\,g_1^2\,g_3^2\,LF_{2,1,0}\big[m_{\tilde d}^{\,i2},m_3\big]\,\delta_{i1i2}-$$
$$\frac{1}{27}\,g_1^2\,g_3^2\,LF_{3,1,-1}\big[m_{\tilde d}^{\,i2},m_3\big]\,\delta_{i1i2}-\frac{1}{9}\,g_1^2\,\overline{a_e}^{pr}\,a_e^{\,pr}\,LF_{2,2,0}\big[m_{\tilde e}^{\,r},m_{\tilde l}^{\,p}\big]\,\delta_{i1i2}+$$
$$\frac{1}{18}\,g_1^2\,\overline{a_e}^{pr}\,a_e^{\,pr}\,LF_{3,2,-1}\big[m_{\tilde e}^{\,r},m_{\tilde l}^{\,p}\big]\,\delta_{i1i2}+\frac{1}{9}\,g_1^2\,\overline{a_e}^{pr}\,a_e^{\,pr}\,LF_{3,1,0}\big[m_{\tilde l}^{\,p},m_{\tilde e}^{\,r}\big]\,\delta_{i1i2}+$$
$$\frac{1}{9}\,g_1^2\,\overline{a_e}^{pr}\,a_e^{\,pr}\,LF_{3,2,-1}\big[m_{\tilde l}^{\,p},m_{\tilde e}^{\,r}\big]\,\delta_{i1i2}-\frac{1}{4}\,g_1^2\,\overline{a_e}^{pr}\,a_e^{\,pr}\,LF_{4,1,-1}\big[m_{\tilde l}^{\,p},m_{\tilde e}^{\,r}\big]\,\delta_{i1i2}+$$
$$\frac{1}{9}\,g_1^2\,\overline{a_e}^{pr}\,a_e^{\,pr}\,LF_{5,1,-2}\big[m_{\tilde l}^{\,p},m_{\tilde e}^{\,r}\big]\,\delta_{i1i2}+\frac{1}{9}\,g_1^2\,\overline{a_d}^{pr}\,a_d^{\,pr}\,LF_{3,1,0}\big[m_{\tilde q}^{\,p},m_{\tilde d}^{\,r}\big]\,\delta_{i1i2}+$$
$$\frac{1}{9}\,g_1^2\,\overline{a_d}^{pr}\,a_d^{\,pr}\,LF_{3,2,-1}\big[m_{\tilde q}^{\,p},m_{\tilde d}^{\,r}\big]\,\delta_{i1i2}-\frac{5}{12}\,g_1^2\,\overline{a_d}^{pr}\,a_d^{\,pr}\,LF_{4,1,-1}\big[m_{\tilde q}^{\,p},m_{\tilde d}^{\,r}\big]\,\delta_{i1i2}+$$
$$\frac{1}{3}\,g_1^2\,\overline{a_d}^{pr}\,a_d^{\,pr}\,LF_{5,1,-2}\big[m_{\tilde q}^{\,p},m_{\tilde d}^{\,r}\big]\,\delta_{i1i2}+\frac{1}{18}\,g_1^2\,\tilde\mu^2\,\overline{y_u}^{pr}\,y_u^{\,pr}\,LF_{2,2,0}\big[m_{\tilde q}^{\,p},m_{\tilde u}^{\,r}\big]\,\delta_{i1i2}-$$
$$\frac{1}{36}\,g_1^2\,\tilde\mu^2\,\overline{y_u}^{pr}\,y_u^{\,pr}\,LF_{3,2,-1}\big[m_{\tilde q}^{\,p},m_{\tilde u}^{\,r}\big]\,\delta_{i1i2}-\frac{1}{18}\,g_1^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{2,1,0}\big[m_{\tilde q}^{\,p},\tilde\mu\big]+$$
$$\frac{1}{36}\,g_1^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{2,2,-1}\big[m_{\tilde q}^{\,p},\tilde\mu\big]+\frac{1}{36}\,g_1^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{3,1,-1}\big[m_{\tilde q}^{\,p},\tilde\mu\big]-$$
$$\frac{1}{18}\,g_1^2\,\tilde\mu^2\,\overline{y_u}^{pr}\,y_u^{\,pr}\,LF_{3,1,0}\big[m_{\tilde u}^{\,r},m_{\tilde q}^{\,p}\big]\,\delta_{i1i2}-\frac{1}{18}\,g_1^2\,\tilde\mu^2\,\overline{y_u}^{pr}\,y_u^{\,pr}\,LF_{3,2,-1}\big[m_{\tilde u}^{\,r},m_{\tilde q}^{\,p}\big]\,\delta_{i1i2}-$$
$$\frac{1}{6}\,g_1^2\,\tilde\mu^2\,\overline{y_u}^{pr}\,y_u^{\,pr}\,LF_{4,1,-1}\big[m_{\tilde u}^{\,r},m_{\tilde q}^{\,p}\big]\,\delta_{i1i2}+\frac{1}{3}\,g_1^2\,\tilde\mu^2\,\overline{y_u}^{pr}\,y_u^{\,pr}\,LF_{5,1,-2}\big[m_{\tilde u}^{\,r},m_{\tilde q}^{\,p}\big]\,\delta_{i1i2}+$$
$$\frac{5}{36}\,g_1^4\,LF_{4,1,-2}[\tilde\mu,m_1]\,\delta_{i1i2}-\frac{1}{9}\,g_1^4\,LF_{5,1,-3}[\tilde\mu,m_1]\,\delta_{i1i2}+\frac{5}{12}\,g_1^2\,g_2^2\,LF_{4,1,-2}[\tilde\mu,m_2]\,\delta_{i1i2}-$$
$$\frac{1}{3}\,g_1^2\,g_2^2\,LF_{5,1,-3}[\tilde\mu,m_2]\,\delta_{i1i2}-\frac{1}{2}\,\overline{y_d}^{ps}\,\overline{y_d}^{ri1}\,y_d^{\,pi2}\,y_d^{\,rs}\,LF_{2,1,0}\big[\tilde\mu,m_{\tilde d}^{\,s}\big]+$$
$$\overline{y_d}^{ps}\,\overline{y_d}^{ri1}\,y_d^{\,pi2}\,y_d^{\,rs}\,LF_{3,1,-1}\big[\tilde\mu,m_{\tilde d}^{\,s}\big]-\frac{1}{2}\,\overline{y_d}^{ps}\,\overline{y_d}^{ri1}\,y_d^{\,pi2}\,y_d^{\,rs}\,LF_{4,1,-2}\big[\tilde\mu,m_{\tilde d}^{\,s}\big]+$$

$$\frac{1}{36}\,g_1^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{2,1,0}\big[\tilde\mu,m_{\tilde q}^{\,p}\big]+\frac{7}{36}\,g_1^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{3,1,-1}\big[\tilde\mu,m_{\tilde q}^{\,p}\big]-$$
$$\frac{1}{18}\,g_1^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{4,1,-2}\big[\tilde\mu,m_{\tilde q}^{\,p}\big]-\frac{1}{2}\,\overline{y_d}^{pi1}\,y_d^{\,ri2}\,\overline{y_u}^{rs}\,y_u^{\,ps}\,LF_{2,1,0}\big[\tilde\mu,m_{\tilde u}^{\,s}\big]+$$
$$\overline{y_d}^{pi1}\,y_d^{\,ri2}\,\overline{y_u}^{rs}\,y_u^{\,ps}\,LF_{3,1,-1}\big[\tilde\mu,m_{\tilde u}^{\,s}\big]-\frac{1}{2}\,\overline{y_d}^{pi1}\,y_d^{\,ri2}\,\overline{y_u}^{rs}\,y_u^{\,ps}\,LF_{4,1,-2}\big[\tilde\mu,m_{\tilde u}^{\,s}\big]-$$
$$\frac{1}{36}\,m_1\,g_1^2\,y_d^{\,pi2}\,\overline{a_d}^{pi1}\,LF_{2,1,1,0}\big[m_1,m_{\tilde d}^{\,i1},m_{\tilde q}^{\,p}\big]+\frac{1}{72}\,m_1\,g_1^2\,y_d^{\,pi2}\,\overline{a_d}^{pi1}\,LF_{2,2,1,-1}\big[m_1,m_{\tilde d}^{\,i1},m_{\tilde q}^{\,p}\big]+$$
$$\frac{1}{36}\,m_1\,g_1^2\,y_d^{\,pi2}\,\overline{a_d}^{pi1}\,LF_{3,1,1,-1}\big[m_1,m_{\tilde d}^{\,i1},m_{\tilde q}^{\,p}\big]-\frac{1}{72}\,g_1^4\,LF_{2,1,1,-1}\big[m_1,m_{\tilde d}^{\,i1},\tilde\mu\big]\,\delta_{i1i2}-$$
$$\frac{1}{36}\,g_1^4\,m_1^2\,LF_{2,1,1,0}\big[m_1,m_{\tilde d}^{\,i1},\tilde\mu\big]\,\delta_{i1i2}-\frac{1}{36}\,m_1\,g_1^2\,\overline{y_d}^{pi1}\,a_d^{\,pi2}\,LF_{2,1,1,0}\big[m_1,m_{\tilde d}^{\,i2},m_{\tilde q}^{\,p}\big]+$$
$$\frac{1}{72}\,m_1\,g_1^2\,\overline{y_d}^{pi1}\,a_d^{\,pi2}\,LF_{2,2,1,-1}\big[m_1,m_{\tilde d}^{\,i2},m_{\tilde q}^{\,p}\big]+\frac{1}{36}\,m_1\,g_1^2\,\overline{y_d}^{pi1}\,a_d^{\,pi2}\,LF_{3,1,1,-1}\big[m_1,m_{\tilde d}^{\,i2},m_{\tilde q}^{\,p}\big]-$$
$$\frac{1}{72}\,g_1^4\,LF_{2,1,1,-1}\big[m_1,m_{\tilde d}^{\,i2},\tilde\mu\big]\,\delta_{i1i2}-\frac{1}{36}\,g_1^4\,m_1^2\,LF_{2,1,1,0}\big[m_1,m_{\tilde d}^{\,i2},\tilde\mu\big]\,\delta_{i1i2}-$$
$$\frac{1}{72}\,m_1\,g_1^2\,y_d^{\,pi2}\,\overline{a_d}^{pi1}\,LF_{2,2,1,-1}\big[m_1,m_{\tilde q}^{\,p},m_{\tilde d}^{\,i1}\big]-\frac{1}{72}\,m_1\,g_1^2\,\overline{y_d}^{pi1}\,a_d^{\,pi2}\,LF_{2,2,1,-1}\big[m_1,m_{\tilde q}^{\,p},m_{\tilde d}^{\,i2}\big]+$$
$$\frac{1}{4}\,g_1^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{2,1,1,-1}\big[m_1,m_{\tilde q}^{\,p},\tilde\mu\big]-\frac{1}{6}\,g_1^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{3,1,1,-2}\big[m_1,m_{\tilde q}^{\,p},\tilde\mu\big]-$$
$$\frac{1}{72}\,g_1^2\,\left(6\,\overline{y_d}^{pi1}\,y_d^{\,pi2}+g_1^2\,\delta_{i1i2}\right)LF_{2,2,1,-2}\big[m_1,\tilde\mu,m_{\tilde d}^{\,i1}\big]+\frac{1}{72}\,g_1^4\,m_1^2$$
$$LF_{2,2,1,-1}\big[m_1,\tilde\mu,m_{\tilde d}^{\,i1}\big]\,\delta_{i1i2}-\frac{1}{72}\,g_1^2\,\left(6\,\overline{y_d}^{pi1}\,y_d^{\,pi2}+g_1^2\,\delta_{i1i2}\right)LF_{2,2,1,-2}\big[m_1,\tilde\mu,m_{\tilde d}^{\,i2}\big]+$$
$$\frac{1}{72}\,g_1^4\,m_1^2\,LF_{2,2,1,-1}\big[m_1,\tilde\mu,m_{\tilde d}^{\,i2}\big]\,\delta_{i1i2}-\frac{5}{24}\,g_1^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{2,2,1,-2}\big[m_1,\tilde\mu,m_{\tilde q}^{\,p}\big]+$$
$$\frac{9}{4}\,g_2^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{2,1,1,-1}\big[m_2,m_{\tilde q}^{\,p},\tilde\mu\big]-\frac{3}{2}\,g_2^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{3,1,1,-2}\big[m_2,m_{\tilde q}^{\,p},\tilde\mu\big]-$$
$$\frac{9}{8}\,g_2^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{2,2,1,-2}\big[m_2,\tilde\mu,m_{\tilde q}^{\,p}\big]+\frac{2}{3}\,m_3\,g_3^2\,y_d^{\,pi2}\,\overline{a_d}^{pi1}\,LF_{2,1,1,0}\big[m_3,m_{\tilde d}^{\,i1},m_{\tilde q}^{\,p}\big]-$$
$$\frac{1}{3}\,m_3\,g_3^2\,y_d^{\,pi2}\,\overline{a_d}^{pi1}\,LF_{2,2,1,-1}\big[m_3,m_{\tilde d}^{\,i1},m_{\tilde q}^{\,p}\big]-\frac{2}{3}\,m_3\,g_3^2\,y_d^{\,pi2}\,\overline{a_d}^{pi1}\,LF_{3,1,1,-1}\big[m_3,m_{\tilde d}^{\,i1},m_{\tilde q}^{\,p}\big]+$$
$$\frac{2}{3}\,m_3\,g_3^2\,\overline{y_d}^{pi1}\,a_d^{\,pi2}\,LF_{2,1,1,0}\big[m_3,m_{\tilde d}^{\,i2},m_{\tilde q}^{\,p}\big]-\frac{1}{3}\,m_3\,g_3^2\,\overline{y_d}^{pi1}\,a_d^{\,pi2}\,LF_{2,2,1,-1}\big[m_3,m_{\tilde d}^{\,i2},m_{\tilde q}^{\,p}\big]-$$
$$\frac{2}{3}\,m_3\,g_3^2\,\overline{y_d}^{pi1}\,a_d^{\,pi2}\,LF_{3,1,1,-1}\big[m_3,m_{\tilde d}^{\,i2},m_{\tilde q}^{\,p}\big]+\frac{1}{3}\,m_3\,g_3^2\,y_d^{\,pi2}\,\overline{a_d}^{pi1}\,LF_{2,2,1,-1}\big[m_3,m_{\tilde q}^{\,p},m_{\tilde d}^{\,i1}\big]+$$
$$\frac{1}{3}\,m_3\,g_3^2\,\overline{y_d}^{pi1}\,a_d^{\,pi2}\,LF_{2,2,1,-1}\big[m_3,m_{\tilde q}^{\,p},m_{\tilde d}^{\,i2}\big]-\frac{1}{8}\,\tilde\mu^2\,\overline{y_d}^{pi1}\,y_d^{\,ri2}\,\overline{y_u}^{rs}\,y_u^{\,ps}\,LF_{2,2,1,-1}\big[m_{\tilde q}^{\,p},\tilde\mu,m_{\tilde u}^{\,s}\big]-$$
$$\frac{1}{8}\,\tilde\mu^2\,\overline{y_d}^{pi1}\,y_d^{\,ri2}\,\overline{y_u}^{rs}\,y_u^{\,ps}\,LF_{2,2,1,-1}\big[m_{\tilde q}^{\,r},\tilde\mu,m_{\tilde u}^{\,s}\big]+$$
$$\frac{1}{8}\,\tilde\mu^2\,\overline{y_d}^{pi1}\,y_d^{\,ri2}\,\overline{y_u}^{rs}\,y_u^{\,ps}\,LF_{2,2,1,-1}\big[m_{\tilde u}^{\,s},\tilde\mu,m_{\tilde q}^{\,p}\big]+$$
$$\frac{1}{8}\,\tilde\mu^2\,\overline{y_d}^{pi1}\,y_d^{\,ri2}\,\overline{y_u}^{rs}\,y_u^{\,ps}\,LF_{2,2,1,-1}\big[m_{\tilde u}^{\,s},\tilde\mu,m_{\tilde q}^{\,r}\big]+\frac{1}{4}\,g_1^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{2,1,1,-1}\big[\tilde\mu,m_1,m_{\tilde d}^{\,i1}\big]-$$
$$\frac{1}{6}\,g_1^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{3,1,1,-2}\big[\tilde\mu,m_1,m_{\tilde d}^{\,i1}\big]+\frac{1}{4}\,g_1^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{2,1,1,-1}\big[\tilde\mu,m_1,m_{\tilde d}^{\,i2}\big]-$$
$$\frac{1}{6}\,g_1^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{3,1,1,-2}\big[\tilde\mu,m_1,m_{\tilde d}^{\,i2}\big]-\frac{1}{8}\,g_1^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{2,1,1,-1}\big[\tilde\mu,m_1,m_{\tilde q}^{\,p}\big]-$$
$$\frac{3}{8}\,g_2^2\,\overline{y_d}^{pi1}\,y_d^{\,pi2}\,LF_{2,1,1,-1}\big[\tilde\mu,m_2,m_{\tilde q}^{\,p}\big]+\frac{1}{4}\,\tilde\mu^2\,\overline{y_d}^{pi1}\,y_d^{\,ri2}\,\overline{y_u}^{rs}\,y_u^{\,ps}\,LF_{2,1,1,0}\big[\tilde\mu,m_{\tilde q}^{\,p},m_{\tilde u}^{\,s}\big]-$$
$$\frac{1}{4}\,\tilde\mu^2\,\overline{y_d}^{pi1}\,y_d^{\,ri2}\,\overline{y_u}^{rs}\,y_u^{\,ps}\,LF_{3,1,1,-1}\big[\tilde\mu,m_{\tilde q}^{\,p},m_{\tilde u}^{\,s}\big]+\frac{1}{4}\,\tilde\mu^2\,\overline{y_d}^{pi1}\,y_d^{\,ri2}\,\overline{y_u}^{rs}\,y_u^{\,ps}$$
$$LF_{2,1,1,0}\big[\tilde\mu,m_{\tilde q}^{\,r},m_{\tilde u}^{\,s}\big]-\frac{1}{4}\,\tilde\mu^2\,\overline{y_d}^{pi1}\,y_d^{\,ri2}\,\overline{y_u}^{rs}\,y_u^{\,ps}\,LF_{3,1,1,-1}\big[\tilde\mu,m_{\tilde q}^{\,r},m_{\tilde u}^{\,s}\big]+\frac{1}{12}\,m_1\,g_1^2\,y_d^{\,pi2}$$
$$\overline{a_d}^{pi1}\,LF_{1,1,1,1,0}\big[m_1,m_{\tilde d}^{\,i1},m_{\tilde q}^{\,p},\tilde\mu\big]+\frac{1}{24}\,m_1\,g_1^2\,y_d^{\,pi2}\,\overline{a_d}^{pi1}\,LF_{2,1,1,1,-1}\big[m_1,m_{\tilde d}^{\,i1},m_{\tilde q}^{\,p},\tilde\mu\big]+$$
$$\frac{1}{12}\,m_1\,g_1^2\,\overline{y_d}^{pi1}\,a_d^{\,pi2}\,LF_{1,1,1,1,0}\big[m_1,m_{\tilde d}^{\,i2},m_{\tilde q}^{\,p},\tilde\mu\big]+\frac{1}{24}\,m_1\,g_1^2\,\overline{y_d}^{pi1}\,a_d^{\,pi2}$$
$$LF_{2,1,1,1,-1}\big[m_1,m_{\tilde d}^{\,i2},m_{\tilde q}^{\,p},\tilde\mu\big]-\frac{1}{4}\,\overline{y_d}^{si1}\,y_d^{\,pi2}\,\overline{a_d}^{pr}\,a_d^{\,sr}\,LF_{2,1,1,1,-1}\big[m_{\tilde d}^{\,r},m_{\tilde q}^{\,p},m_{\tilde q}^{\,s},\tilde\mu\big]+$$
$$\frac{1}{24}\,m_1\,g_1^2\,y_d^{\,pi2}\,\overline{a_d}^{pi1}\,LF_{2,1,1,1,-1}\big[m_{\tilde d}^{\,i1},m_1,m_{\tilde q}^{\,p},\tilde\mu\big]+\frac{1}{24}\,m_1\,g_1^2\,\overline{y_d}^{pi1}\,a_d^{\,pi2}$$
$$LF_{2,1,1,1,-1}\big[m_{\tilde d}^{\,i2},m_1,m_{\tilde q}^{\,p},\tilde\mu\big]+\frac{1}{18}\,g_1^2\,\overline{a_d}^{pi1}\,a_d^{\,pi2}\,LF_{2,1,1,1,-1}\big[m_{\tilde q}^{\,p},m_1,m_{\tilde d}^{\,i1},m_{\tilde d}^{\,i2}\big]-$$
$$\frac{1}{24}\,m_1\,g_1^2\,y_d^{\,pi2}\,\overline{a_d}^{pi1}\,LF_{2,1,1,1,-1}\big[m_{\tilde q}^{\,p},m_1,m_{\tilde d}^{\,i1},\tilde\mu\big]-\frac{1}{24}\,m_1\,g_1^2\,\overline{y_d}^{pi1}\,a_d^{\,pi2}$$
$$LF_{2,1,1,1,-1}\big[m_{\tilde q}^{\,p},m_1,m_{\tilde d}^{\,i2},\tilde\mu\big]+\frac{2}{3}\,g_3^2\,\overline{a_d}^{pi1}\,a_d^{\,pi2}\,LF_{2,1,1,1,-1}\big[m_{\tilde q}^{\,p},m_3,m_{\tilde d}^{\,i1},m_{\tilde d}^{\,i2}\big]+$$
$$\frac{1}{4}\,\tilde\mu^2\,\overline{y_d}^{pi1}\,y_d^{\,ri2}\,\overline{y_u}^{rs}\,y_u^{\,ps}\,LF_{2,1,1,1,-1}\big[m_{\tilde u}^{\,s},m_{\tilde q}^{\,p},m_{\tilde q}^{\,r},\tilde\mu\big]-\frac{1}{24}\,m_1\,g_1^2\,y_d^{\,pi2}\,\overline{a_d}^{pi1}$$
$$LF_{2,1,1,1,-1}\big[\tilde\mu,m_1,m_{\tilde d}^{\,i1},m_{\tilde q}^{\,p}\big]-\frac{1}{24}\,m_1\,g_1^2\,\overline{y_d}^{pi1}\,a_d^{\,pi2}\,LF_{2,1,1,1,-1}\big[\tilde\mu,m_1,m_{\tilde d}^{\,i2},m_{\tilde q}^{\,p}\big]\Big)$$